Shikhar Bharadwaj

PhD Student, **Carnegie Mellon University**

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Education

Carnegie Mellon University — Pittsburgh, PA, USA
Advisors: David Mortensen and Shinji Watanabe — *August 2024 - Present*

Indian Institute of Science — Bengaluru, India
M.Tech (Research) in Intelligent Systems — *August 2019 - June 2022*

Birla Institute of Technology and Science — Hyderabad, India
B.E. (Honors) in Computer Science Engineering — *August 2014 - June 2018*

Research Experience

Meta Reality Labs — USA
Research Scientist Intern — *May 2026 - August 2026*

Mitsubishi Electric Research Laboratories — Boston, USA
PhD Research Intern — *May 2025 - August 2025*

Google DeepMind — Bengaluru, India
Pre-Doctoral Researcher — *May 2022 - July 2024*

Microsoft Research — Bengaluru, India
Research Intern — *December 2017 - May 2018*

Selected Manuscripts

* denotes equal contribution

[1] **Phone Segmentation and Recognition through Phonological Activation Mapping**
Shikhar Bharadwaj, Kwanghee Choi, Stephen McIntosh, Chin-Jou Li, Eunjung Yeo, Daisuke Saito, Nobuaki Minematsu, Shinji Watanabe, Jian Zhu, David Harwath, David Mortensen
Preprint 2026

[2] **An Empirical Recipe for Universal Phone Recognition**
Shikhar Bharadwaj, Chin-Jou Li, Kwanghee Choi, Eunjung Yeo, William Chen, Shinji Watanabe, David Mortensen
Interspeech 2026 (Oral)

[3] **The CMU-AIST Submission for the ICME 2025 Audio Encoder Challenge**
Shikhar Bharadwaj*, Samuele Cornell*, Kwanghee Choi, Hye-jin Shim, Soham Deshmukh, Satoru Fukayama, Shinji Watanabe
Technical Report 2026

[4] **PRiSM: Benchmarking Phone Realization in Speech Models**
Shikhar Bharadwaj*, Chin-Jou Li*, Yoonjae Kim*, Kwanghee Choi, Eunjung Yeo, Ryan Shim, Hanyu Zhou, Brendon Boldt, Karen Jacome, Kalvin Chang, Darsh Agrawal, Keer Xu, Huck Yang, Jian Zhu, Shinji Watanabe, David Mortensen
ACL 2026

[5] **POWSM: A Phonetic Open Whisper-Style Speech Foundation Model**
Chin-Jou Li*, Kalvin Chang*, Shikhar Bharadwaj, Eunjung Yeo, Kwanghee Choi, Jian Zhu, David Mortensen, Shinji Watanabe
ACL 2026 (Oral)

[6] **OpenBEATs: A Fully Open-Source General-Purpose Audio Encoder**
Shikhar Bharadwaj, Samuele Cornell, Kwanghee Choi, Satoru Fukayama, Hye-jin Shim, Soham Deshmukh, Shinji Watanabe
WASPAA 2025 (Oral)

[7] **Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities**

Gemini Team, Google (including Shikhar Bharadwaj)
Technical Report 2025

[8] **OpusLM: A Family of Open Unified Speech Language Models**
Jinchuan Tian, William Chen, Yifan Peng, Jiatong Shi, Siddhant Arora, Shikhar Bharadwaj, Takashi Maekaku, Yusuke Shinohara, Keita Goto, Xiang Yue, Huck Yang, Shinji Watanabe
Interspeech 2025

[9] **IndicGenBench: A Multilingual Benchmark to Evaluate Generation Capabilities of LLMs on Indic Languages**
Harman Singh, Nitish Gupta, Shikhar Bharadwaj, Dinesh Tewari, Partha Talukdar
ACL 2024

[10] **STAB: Speech Tokenizer Assessment Benchmark**
Shikhar Vashishth*, Harman Singh*, Shikhar Bharadwaj*, Sriram Ganapathy, Chulayuth Asawaroengchai, Kartik Audhkhasi, Andrew Rosenberg, Ankur Bapna, Bhuvana Ramabhadran
Preprint 2024

[11] **Multimodal Modeling for Spoken Language Identification**
Shikhar Bharadwaj*, Min Ma*, Shikhar Vashishth*, Ankur Bapna, Sriram Ganapathy, Vera Axelrod, Siddharth Dalmia, Wei Han, Yu Zhang, Daan van Esch, Sandy Ritchie, Partha Talukdar, Jason Riesa
ICASSP 2024

Selected Research Directions

- **Audio Representation Learning**
  - Trained OpenBEATs [6] , a masked audio token prediction based audio encoder.
  - By scaling pre-training to 70k hours of open-source speech and audio, OpenBEATs outperforms billion-parameter MAE models using only 25% of their parameter count.
  - Our submission based on OpenBEATs was the top performing self-supervised model in the ICME 2025 Audio Encoder Challenge, and achieved third place overall.

- **Multilingual Speech Representations**
  - Created PRiSM [4] to benchmark the phonetic capabilities of speech models, focusing on phone recognition in unseen languages and downstream low-resource application areas such as accented, multilingual, and pathological speech.
  - Developed POWSM [5], a phonetic open Whisper-style speech foundation model that unifies multiple phone-related tasks including ASR, phone recognition, grapheme-to-phoneme and phoneme-to-grapheme.
  - Created MuSeLi [11] to jointly model speech and text for language identification at the scale of YouTube. This is the current SOTA for language identification and received several awards at Google.

Professional Experience

Myntra Designs Pvt Ltd. — Remote, India
Machine Learning Intern — *April 2020 - June 2020*

Media.net — Mumbai, India
Software Engineer — *July 2018 - July 2019*

Awards and Recognition

- **Google Internal Awards**: Five awards for contributions towards Google4India event, Google-USM and multimodal language identification work [11]
- **Media Coverage**: Audio collection effort in collaboration with the *Indian Institute of Science* has been covered by The White House and The Economic Times
- **Academic Excellence**: Institute Merit Scholarship (Top 2% for 7 semesters, Top 1% for 1 semester), NTSE Scholarship, INSPIRE Scholarship